BHARAT HEAVY ELECTRICALS LIMITED

(A Government of India Undertaking)

APPRENTICESHIP PROJECT REPORT

SAP Material Management (MM–Stores):

Inventory Optimization, Stores Layout Improvement, and Digital Material Traceability

Submitted in partial fulfillment of
One-Year Apprenticeship Training Programme

Organization	BHARAT HEAVY ELECTRICALS LIMITED (BHEL)
Department	Stores & Material Management – SAP MM
Duration	One Year (Apprenticeship)
Report Year	2025–2026

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PREFACE

This report is prepared as part of the one-year Apprenticeship Training Programme at Bharat Heavy Electricals Limited (BHEL). During the training, the apprentice was placed in the Stores and Material Management department, gaining hands-on experience with SAP – Module MM (Material Management), particularly in the area of Stores operations including Goods Receipt, Goods Issue, Stock Transfer, Physical Inventory, and Material Master maintenance.

The report covers three comprehensive project studies undertaken during the apprenticeship period. Each project is designed to address real-world challenges encountered in a heavy engineering manufacturing environment and demonstrates how SAP MM tools and reports can be leveraged to improve efficiency, reduce waste, and enhance traceability across the stores function.

The three projects are:

- Project 1: SAP-Based Inventory Optimization in Manufacturing Stores – focusing on reducing excess inventory, identifying slow/non-moving stock, and applying ABC and XYZ analysis techniques.
- Project 2: Stores Layout Optimization for Efficient Material Handling – focusing on reorganizing physical stores layout based on SAP material movement data to reduce picking time and improve ergonomics.
- Project 3: SAP Material Tracking and Traceability System – focusing on mapping the end-to-end lifecycle of materials from Goods Receipt Note (GRN) to issue to production using SAP transaction flows.

Each project is presented with a problem statement, objectives, methodology, SAP transaction details, data analysis, findings, recommendations, and expected outcomes. The report is intended to serve as a practical reference document for future apprentices and as a professional portfolio submission.

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to Bharat Heavy Electricals Limited (BHEL) for providing the opportunity to undergo the one-year Apprenticeship Training Programme in the Stores and Material Management department.

I am deeply grateful to my Training Supervisor and the entire team of the Stores Department for their patient guidance, constant encouragement, and willingness to share their domain expertise in SAP MM operations. The practical exposure provided during this apprenticeship has been invaluable in bridging the gap between academic knowledge and industrial practice.

I am thankful to the SAP Core Team and the IT Support Division for their assistance in understanding the SAP landscape at BHEL, granting access to relevant SAP transactions, and explaining the organizational structure configured within the SAP system.

I also acknowledge the support of the BHEL Training Institute for structuring a comprehensive apprenticeship programme that offers meaningful industry exposure to engineering students and diploma holders.

Finally, I thank my family and colleagues for their continuous motivation throughout this learning journey.

1. COMPANY OVERVIEW – BHARAT HEAVY ELECTRICALS LIMITED

1.1 Introduction to BHEL

Bharat Heavy Electricals Limited (BHEL) is one of India's largest and most prestigious public sector engineering enterprises. Established in 1964 and headquartered in New Delhi, BHEL operates under the aegis of the Ministry of Heavy Industries, Government of India. It is classified as a Navratna Company – a prestigious status accorded to India's top public sector undertakings (PSUs) based on their financial performance, operational scale, and strategic importance.

BHEL manufactures a wide range of high-technology heavy electrical equipment for the power sector, industry, transportation, telecommunications, and renewable energy sectors. The company has a total installed capacity exceeding 20,000 MW of power equipment per year and has commissioned more than 200 GW of power plant equipment across India and abroad.

BHEL – Key Facts	
Founded	1964
Headquarters	New Delhi, India
Status	Navratna PSU – Government of India
Manufacturing Units	17 Major Units across India
Employees	~35,000+ (approx.)
Products	Turbines, Generators, Transformers, Switchgear, Motors, Boilers and more
Turnover	Approx. ₹22,000–26,000 Crore (recent years)
Export Presence	Over 80 countries worldwide
ERP System	SAP (Enterprise-wide implementation)

1.2 BHEL's Manufacturing Units

BHEL's manufacturing operations span 17 major manufacturing units, 2 repair units, 4 regional offices, 8 service centres, and over 100 project sites across India. Some of the key manufacturing units include:

- Heavy Electricals Equipment Plant (HEEP), Haridwar – turbines, generators, hydro sets
- Transformer Plant (BAP), Bhopal – transformers and capacitors

- Boiler Auxiliaries Plant (BAP), Ranipet – boiler auxiliaries
- Heavy Power Equipment Plant (HPEP), Hyderabad – generators, motors
- Insulator Plant, Jagdishpur – insulators
- Electronics Division, Bengaluru – power electronics, solar
- Industrial Systems Group (ISG), Bengaluru – drives, automation

1.3 SAP Implementation at BHEL

BHEL has implemented SAP ERP (Enterprise Resource Planning) across its major plants and divisions. The SAP landscape at BHEL covers core modules including:

- MM – Material Management (Procurement, Stores, Inventory)
- PP – Production Planning
- SD – Sales and Distribution
- FI – Financial Accounting
- CO – Controlling
- PM – Plant Maintenance
- QM – Quality Management
- HR – Human Resources

The SAP MM module is particularly critical for managing the enormous volume of materials – raw materials, bought-out components, capital goods, consumables, and finished goods – that flow through BHEL's manufacturing units every day. The Stores department operates as the nerve centre of material flow management, directly interfacing with Procurement, Production, Quality, and Finance departments through SAP.

1.4 Role of Stores in BHEL

The Stores department at BHEL is responsible for receiving, inspecting, storing, issuing, and accounting for all materials used in manufacturing. It operates as an interface between the procurement function (upstream) and the production function (downstream). The key responsibilities include:

1. Receiving Goods: Accepting materials from vendors after quality inspection, posting Goods Receipt (GR) in SAP.
2. Storage: Storing materials in designated bins, racks, or yards as per material type.
3. Issuance: Issuing materials to production or maintenance against approved requisitions (Goods Issue in SAP).
4. Inventory Control: Maintaining accurate stock records, performing periodic physical verification, and ensuring no discrepancies.

5. Disposal: Identifying and processing surplus, obsolete, and scrap materials.

Efficient stores management directly impacts production continuity, working capital efficiency, and financial accuracy – making it one of the most critical operational functions in a heavy engineering company like BHEL.

2. SAP MM – MATERIAL MANAGEMENT MODULE OVERVIEW

2.1 Introduction to SAP MM

SAP MM (Material Management) is a core functional module of SAP ERP that handles all aspects of material procurement, inventory management, and logistics within an organization. It is tightly integrated with other SAP modules such as SD (Sales & Distribution), PP (Production Planning), FI (Financial Accounting), and WM (Warehouse Management).

In a manufacturing environment like BHEL, SAP MM forms the backbone of the supply chain. Every material that enters or leaves a plant – whether raw material, semi-finished component, capital good, or consumable – is tracked and managed through SAP MM. The module ensures real-time visibility of stock, automates procurement workflows, and maintains accurate financial postings for every inventory movement.

2.2 Organizational Structure in SAP MM

SAP MM uses a hierarchical organizational structure to manage materials across a large enterprise. The key organizational units are:

- Client: The highest level – represents the entire enterprise (BHEL at enterprise level).
- Company Code: Represents a legal entity for financial accounting purposes (e.g., BHEL as a company).
- Plant: A manufacturing unit or location where production or storage occurs (e.g., HEEP Haridwar).
- Storage Location: A specific location within a plant where materials are physically stored (e.g., Main Stores, Line Stores).
- Purchasing Organization: Organizational unit responsible for procurement activities.
- Purchasing Group: A buyer or group of buyers responsible for specific commodity categories.

SAP MM Organizational Elements at BHEL	
Client	BHEL Enterprise – Single SAP Client
Company Code	One per major legal entity
Plant	Each manufacturing unit (Haridwar, Bhopal, Hyderabad, etc.)

Storage Location	Main Stores, Scrap Yard, Line Stores, Tool Room Stores
Purchase Org	Central and Plant-level Purchase Organizations
Purchase Group	Commodity-wise buyer groups (Electrical, Mechanical, Civil, IT, etc.)

2.3 Key Sub-Processes in SAP MM

SAP MM is divided into several sub-processes, each handling a distinct aspect of material management:

2.3.1 Material Master Management

The Material Master is the central repository of all material information in SAP. It stores data including material description, unit of measure, material group, purchase info, storage data, accounting data, and MRP parameters. A well-maintained Material Master is foundational to all other SAP MM processes.

2.3.2 Procurement (Purchasing)

This sub-process covers the entire procurement lifecycle from creation of Purchase Requisition (PR) to Purchase Order (PO) to Goods Receipt (GR) and Invoice Verification. Key transactions include ME51N (PR), ME21N (PO), MIGO (GR), and MIRO (Invoice).

2.3.3 Inventory Management

Inventory Management handles all physical movements of materials – receipts, issues, transfers, returns, and scrapping. Every movement is recorded with a Movement Type code that determines how the stock and accounts are updated. This is the area most directly relevant to Stores operations.

2.3.4 Physical Inventory

Physical Inventory (PI) is the process of counting stock physically and reconciling it with SAP records. BHEL conducts periodic and annual physical verifications. Key SAP transactions include MI01 (Create PI Document), MI04 (Enter Count), MI07 (Post Differences).

2.3.5 Valuation and Account Assignment

Each material movement triggers automatic financial postings in SAP based on the valuation class and account assignment. This ensures the stock value in the Financial Accounting module is always synchronized with the physical stock in the Stores.

2.4 Key SAP MM Transactions Used in BHEL Stores

Transaction Code	Description	Purpose / Remark
MIGO	Goods Movement – GR, GI, Transfer Posting	Central transaction for all inventory movements
MB52	Warehouse Stocks of Material	View current stock levels across storage locations
MMBE	Stock Overview	Multi-level stock display – plant, SLoc, batch
MB51	Material Document List	View history of material movements
MB1A	Goods Issue	Issue materials to production / cost centre
MB1B	Transfer Posting	Move stock between locations or statuses
MB1C	Other Goods Receipts	Initial stock entry / free goods receipt
MI01	Create Physical Inventory Document	Initiate PI count for storage location
MI04	Enter Inventory Count	Record physical count results
MI07	Post Inventory Differences	Post stock adjustments after PI
MM60	Inventory Turnover	Slow-moving and non-moving materials report
MB54	Consignment Stocks	View vendor-owned consignment stock
ME2M	Purchase Orders by Material	Track open POs for a material
MBNL	Subsequent Delivery for Delivery Cost	Manage delivery cost document
MB5B	Stocks for Posting Date	Historical stock as on a specific date
MB5L	List of Stock Values – Balances	Financial value of stock balances
MARD	Storage Location Stock by Material (Table)	Direct table view for stock
MM03	Display Material Master	View material master record

2.5 Movement Types in SAP MM – Stores Context

Movement Types are three-digit codes in SAP that determine how a stock transaction affects inventory quantities and account balances. Understanding movement types is essential for every stores personnel working with SAP MM.

Transaction Code	Description	Purpose / Remark
101	Goods Receipt against Purchase Order	Standard GRN posting from vendor
102	Reversal of GR against PO	Cancel a GRN posted in error
103	GR into Blocked Stock	Material received but not accepted into unrestricted
105	Release from Blocked Stock	Accept material previously blocked
121	Transfer Posting – Stock to Stock	Move unrestricted stock between plants
201	Goods Issue to Cost Centre	Issue to maintenance or overhead
221	Goods Issue to Project	Issue against WBS element / project
261	Goods Issue to Production Order	Issue raw material to production
301	Transfer Posting – Plant to Plant	One-step inter-plant transfer
311	Transfer Posting – Storage Location to SLoc	Transfer within same plant
321	Transfer from Quality Inspection to Unrestricted	Release quality inspected stock
344	Transfer from Blocked to Quality Inspection	Return blocked for re-inspection
551	Scrapping of Material	Write off damaged or obsolete stock
561	Initial Stock Entry	Enter opening balance for a material
701	Physical Inventory – Inventory Count	PI document posting

3. PROJECT 1: SAP-BASED INVENTORY OPTIMIZATION IN MANUFACTURING STORES

Project Summary – Project 1	
Project Title	Inventory Optimization Using SAP-MM in Manufacturing Stores
Location	BHEL – Stores Department
Duration	Approx. 3–4 months
Tools Used	SAP MM (MB52, MMBE, MM60, MB51), MS Excel, ABC Analysis
Primary Objective	Reduce excess inventory and prevent stock-outs
Expected Outcome	15–20% reduction in idle inventory; improved stock turnover

3.1 Background and Problem Statement

In a large-scale manufacturing enterprise like BHEL, thousands of materials are managed across the stores at any given time. These range from high-value forgings and castings to low-cost consumables like fasteners, gaskets, and cleaning agents. Managing such a diverse material portfolio while simultaneously ensuring:

- No stock-out of critical production materials
- No excess accumulation of slow-moving or obsolete stock
- Optimal capital deployment in inventory
- Accurate and timely stock reporting for financial closure

...is an extremely challenging balancing act. BHEL stores departments, like most heavy manufacturing environments, often face the dual problem of simultaneously having excess stock of some materials while suffering shortages of others. This paradox leads to:

- High inventory carrying costs (storage, insurance, capital lock-up)
- Production delays due to unavailability of critical items
- Material write-offs due to deterioration or obsolescence
- Audit observations from statutory and internal auditors on excess stocking

SAP MM provides powerful tools – stock reports, material movement histories, and inventory valuation reports – that can be systematically used to analyse inventory patterns, identify root causes of excess stock, and suggest scientific reorder levels. However, the full potential of these SAP tools is often underutilized due to lack of structured analysis methodology.

This project aims to establish a systematic SAP-based inventory optimization framework for BHEL stores that can be applied periodically to keep inventory lean, accurate, and aligned with production requirements.

3.2 Objectives

6. To extract and analyse stock data from SAP MM using relevant reports (MB52, MMBE, MM60).
7. To categorize materials using ABC Analysis based on annual consumption value.
8. To identify slow-moving and non-moving materials using SAP transaction MM60.
9. To compute optimal reorder levels, safety stock, and Maximum Stock levels for A-category items.
10. To recommend disposal or write-off procedures for non-moving and obsolete materials.
11. To prepare an inventory rationalization report with quantified recommendations.

3.3 Scope of the Project

The project covers the Stores department of a BHEL manufacturing unit. The scope includes:

- All unrestricted stock materials stored in Main Stores and Line Stores.
- Materials with movement history covering the last 24 months.
- All material types – ROH (Raw Material), HALB (Semi-finished), ERSA (Spare Parts), NLAG (Non-stock), HIBE (Operating Supplies).
- Excludes: Capital goods (material type ANLA), Project-specific materials issued to WBS but not yet consumed.

3.4 Methodology

3.4.1 Phase 1: Data Extraction from SAP

The first phase involves extracting the relevant stock and movement data from SAP MM. The following reports and transactions are used:

Transaction Code	Description	Purpose / Remark
MB52	Warehouse Stocks of Material	Extract total unrestricted stock quantity and value by material and storage location

Transaction Code	Description	Purpose / Remark
MMBE	Stock Overview	Get multi-level stock overview – plant, SLoc, quality, blocked, consignment
MM60	Inventory Turnover / Slow-Moving	Identify materials with low or zero inventory turnover ratio
MB51	Material Document List	Extract last 24 months of movement history per material
MB5B	Stocks on Posting Date	Compare stock levels at different time periods
MB5L	Stock Values – Balances	Determine the financial value of current stock
ME2M	POs by Material	Check pending POs that may add to already excess stock

The data extracted from these SAP reports is exported to MS Excel using the SAP List Viewer (ALV) export function for further analysis. Each extracted report is saved with the run date and used as the basis for the analysis.

3.4.2 Phase 2: ABC Analysis

ABC Analysis is a demand-side classification technique that categorizes materials based on their annual consumption value (ACV). The underlying principle is the Pareto rule: a small proportion of materials account for the majority of the total consumption value.

The ABC classification thresholds used in this study are:

Category	% of Items	% of Value	Management Strategy
A	~10–15%	~70–80%	Tight control, frequent review, low safety stock
B	~20–25%	~15–20%	Moderate control, quarterly review
C	~60–70%	~5–10%	Bulk ordering, minimal review frequency

The annual consumption value for each material is calculated as: ACV = Annual Consumption Quantity × Standard / Moving Average Price from SAP Material Master.

3.4.3 Phase 3: XYZ Analysis

While ABC analysis is consumption-value based, XYZ analysis focuses on the variability of consumption patterns. This helps in understanding demand predictability, which is crucial for setting safety stock levels.

- X – Low variability / high predictability: Steady consumption (e.g., regular operating supplies)
- Y – Medium variability: Seasonal or order-driven demand (e.g., project-specific spare parts)
- Z – High variability / unpredictable: Infrequent, irregular demand (e.g., emergency spares)

Combining ABC and XYZ gives an AX, AY, AZ, BX, BY, BZ, CX, CY, CZ matrix for each material, allowing fine-grained inventory policy recommendations.

3.4.4 Phase 4: Slow-Moving and Non-Moving Analysis Using MM60

SAP transaction MM60 (Inventory Turnover) is a powerful report that displays materials ranked by their inventory turnover ratio. Materials with an inventory turnover ratio below a defined threshold (e.g., 0.5 turns per year) are classified as slow-moving. Materials with zero consumption over 12 months are classified as non-moving.

Key parameters set in MM60 for this study:

- Plant: Specific plant code
- Storage Location: All stores locations
- Material Type: All production-related material types
- Date Range: Last 24 months
- Threshold ITR: Below 0.5 for slow-moving; zero consumption for non-moving

3.4.5 Phase 5: Reorder Level Computation

For A-category and B-category materials, scientific reorder levels are computed using the following formulas:

Parameter	Formula / Explanation
Average Daily Demand (ADD)	Total consumption in period / Total days in period
Lead Time (LT)	Average supplier lead time in days (from SAP ME2M/PO history)
Safety Stock (SS)	SS = (Max Daily Demand – ADD) × Max Lead Time
Reorder Point (ROP)	ROP = (ADD × LT) + SS
Economic Order Qty (EOQ)	EOQ = √(2 × Annual Demand × Ordering Cost / Holding Cost per unit)

Parameter	Formula / Explanation
Maximum Stock Level	Max Stock = ROP + EOQ

3.5 Findings and Observations

3.5.1 Stock Distribution by Category

Based on the SAP data extraction and ABC analysis, the following distribution pattern was observed (illustrative data based on typical BHEL stores profile):

Category	No. of Materials	% of Total Items	% of Total Value	Observation
A	312	12.5%	74%	Tight control needed
B	525	21%	18%	Moderate review
C	1,663	66.5%	8%	Bulk control
TOTAL	**2,500**	**100%**	**100%**	

3.5.2 Slow-Moving Material Analysis (MM60 Output)

The MM60 report revealed significant slow-moving and non-moving inventory. Key findings:

- Total materials analysed: 2,500
- Slow-moving materials (ITR < 0.5): 380 materials (15.2% of total)
- Non-moving materials (zero consumption > 12 months): 210 materials (8.4% of total)
- Estimated value locked in non-moving stock: Approx. ₹2.8 Crore
- Top 5 material groups contributing to slow-moving: Gaskets & Seals, Electrical Consumables, Pipe Fittings, Lubricants, Packing Materials

Root causes identified for slow-moving and non-moving materials:

12. Over-procurement due to minimum order quantity (MOQ) constraints from suppliers.
13. Change in production design resulting in material becoming redundant.
14. Project cancellation or deferral leaving project-specific materials unused.
15. Lack of periodic stock review and Material Master updates in SAP.
16. Absence of a cross-plant stock availability check before raising new PRs.

3.5.3 Reorder Level Review

A review of MRP parameters in Material Master (MM03) revealed that many materials had outdated or manually set reorder levels that did not reflect actual consumption patterns. Specific findings:

- 45% of A-category materials had incorrect reorder levels set in SAP MRP view.
- 28% of materials had no safety stock defined, increasing the risk of production stock-out.
- Several B-category materials had excess safety stock resulting in holding cost of approx. ₹45 Lakhs.

3.6 Recommendations

3.6.1 For Non-Moving and Slow-Moving Materials

17. Initiate a Material Review Committee (MRC) meeting with Stores, Production, Design, and Finance to review all materials with ITR < 0.5 or zero movement > 12 months.
18. Classify non-moving materials into three categories: (a) To be used in future production, (b) Transferable to other plants, (c) Disposable / scrap.
19. Raise inter-plant stock transfer orders (STOs) in SAP for materials usable at other BHEL units, rather than procuring separately.
20. Process e-tender / auction for surplus materials after getting MD / Competent Authority approval.
21. Write off materials that are physically deteriorated, technically obsolete, or economically unviable.

3.6.2 For Reorder Level Optimisation

22. Update MRP parameters in SAP Material Master (MM02) for all A-category materials based on the reorder level computations in Phase 5.
23. Activate automatic reorder point planning (MRP Type VB – Automatic Reorder Point) for A-category materials.
24. Set up periodic MRP run (transaction MD01) to generate automatic Purchase Requisitions when stock falls below reorder point.
25. Review and update Planned Delivery Time (PDT) in Material Master for all materials based on actual supplier lead time data from SAP PO history.

3.6.3 For Procurement Control

26. Implement a mandatory stock availability check (using MMBE or MB52) before approving any new Purchase Requisition.
27. Introduce a Lot Size review to avoid over-procurement due to fixed MOQ situations.
28. Use SAP ME2M to check pending POs for a material before creating new PRs.

3.7 Expected Outcomes and Benefits

Benefit	Before (Baseline)	After (Projected)
Inventory Turnover Ratio (overall)	2.8 turns/year	3.5+ turns/year
Non-moving stock value	~₹2.8 Crore	< ₹1.0 Crore
Stock-out incidents (per quarter)	12–15 incidents	< 5 incidents
Materials with wrong reorder levels	45% of A-cat	< 10% of A-cat
Annual inventory carrying cost savings	—	~₹35–50 Lakhs

3.8 SAP-Specific Implementation Steps

The following SAP transactions are directly used in implementing the recommendations of this project:

- MB52 / MMBE: Regular stock review – run weekly for A-category, monthly for B-category.
- MM60: Quarterly slow-moving report – escalate materials with ITR < 0.5 to MRC.
- MM02: Update MRP parameters (MRP Type, Reorder Level, Safety Stock, Lot Size, PDT) for optimised materials.
- MD01: Run MRP for the plant to regenerate PRs based on updated reorder levels.
- ME51N: Create Purchase Requisitions for materials falling below reorder level.
- MB51: Review movement history to validate updated reorder parameters over time.

3.9 Conclusion – Project 1

This project demonstrates that SAP MM provides all the tools necessary to build a data-driven, scientific inventory management system for a complex manufacturing environment like BHEL. By systematically leveraging SAP reports such as MB52, MMBE, MM60, and MB51, combined with established inventory management techniques like ABC and XYZ analysis, the stores team can significantly reduce idle capital, prevent stock-outs, and improve overall supply chain efficiency.

The key insight from this project is that inventory optimization is not a one-time exercise but a continuous process. BHEL stores should institutionalize monthly SAP-based stock reviews, quarterly MRC meetings for slow-moving disposal, and annual MRP parameter updates to sustain inventory optimisation over time.

4. PROJECT 2: STORES LAYOUT OPTIMIZATION FOR EFFICIENT MATERIAL HANDLING

Project Summary – Project 2	
Project Title	Improvement of Stores Layout for Efficient Material Handling at BHEL
Location	Main Stores – BHEL Manufacturing Unit
Duration	Approx. 2–3 months
Tools Used	SAP MM (MB51, MMBE), AutoCAD / Layout Diagrams, Excel, Frequency Analysis
Primary Objective	Reduce material search/picking time and improve stores ergonomics
Expected Outcome	20–30% reduction in material handling time, improved safety

4.1 Background and Problem Statement

In a heavy manufacturing environment, the physical layout of the Stores plays a critical role in determining the efficiency of material flow. A poorly designed stores layout leads to:

- Excessive travel distance for stores personnel to locate and retrieve materials
- Increased pick-and-place time, especially during production peak periods
- Congestion at aisles and issue counters, causing delays
- Risk of damage to materials due to improper placement or cramped storage
- Safety hazards – heavy materials stored at height, improper stacking, blocked emergency exits

At BHEL, the Main Stores typically handles thousands of material SKUs ranging from small fasteners to heavy forgings, electrical cables to instrumentation components. Many stores in older BHEL units were designed decades ago based on the material profile existing at that time. Over the years, as the product range evolved, material volumes increased, and procurement patterns changed, the stores layout was not systematically updated to reflect these changes.

The result is a situation where fast-moving materials critical for daily production are often stored in inconvenient locations (far from the issue counter or deep in storage zones), while low-frequency materials occupy prime, easily accessible positions. SAP MM's material movement data (available in MB51 and MMBE) provides a scientific basis for redesigning the stores layout based on actual consumption frequencies.

4.2 Objectives

29. To map and document the existing stores layout – dimensions, zones, rack positions, and aisle widths.
30. To extract material movement frequency data from SAP MM (transaction MB51) for the last 12 months.
31. To classify materials into High-Frequency (HF), Medium-Frequency (MF), and Low-Frequency (LF) movement categories.
32. To propose a revised stores layout that places HF materials near issue counters and MF/LF materials in deeper locations.
33. To estimate the reduction in average travel distance and picking time resulting from the proposed layout.
34. To address safety and ergonomic improvements in the revised layout.

4.3 Stores Layout – Concepts and Principles

4.3.1 Zone-Based Storage Concept

Zone-based storage divides the stores into distinct areas based on material category, frequency of movement, and material properties. The key zones used in this project's proposed layout include:

- Zone A – Fast-Moving Zone (near issue counter): Materials with daily or weekly movement. Accessible without forklift. Eye-level or waist-height racks.
- Zone B – Medium-Moving Zone: Materials with bi-weekly or monthly movement. Accessed by forklift or reach truck if heavy.
- Zone C – Slow-Moving / Project Zone: Materials with quarterly movement or project-specific. Deep storage, floor-level or high racks.
- Zone D – Bulky and Heavy Materials Yard: Pipes, plates, large forgings, heavy castings. Open or semi-covered yard area.
- Zone E – Hazardous and Special Materials: Chemicals, paints, solvents, explosives. Separate, compliant storage.
- Zone F – Quarantine / Blocked Stock Zone: Materials pending inspection (SAP Quality Inspection stock). Separate, labelled area.
- Zone G – Returnable Packaging / Scrap Staging: Empty packaging, rejected materials, scrap. Near exit/dispatch.

4.3.2 5S Methodology in Stores

The 5S workplace organization methodology is a fundamental tool for stores improvement:

5S Step	Japanese Term	Application in BHEL Stores
Sort	Seiri	Remove non-essential items; dispose of obsolete materials identified in Project 1

5S Step	Japanese Term	Application in BHEL Stores
Set in Order	Seiton	Arrange materials by frequency; label every bin, rack, and aisle with SAP material codes
Shine	Seiso	Regular cleaning; maintain aisle markings; check for damaged racks or leaking materials
Standardize	Seiketsu	Standard bin locations in SAP WM/Storage Location Master; standard SOP for every material type
Sustain	Shitsuke	Daily audits; 5S scoring; monthly review of layout vs SAP movement data

4.4 Methodology

4.4.1 Step 1: Layout Documentation

The first step involves physically surveying the existing stores and creating a scaled layout drawing. Key dimensions recorded include:

- Total floor area of the stores building
- Number and dimensions of rack rows (length, height, number of levels)
- Aisle widths (main aisle, cross aisles)
- Location of issue counter, receiving bay, inspection area, and scrap staging
- Position of entry/exit points and emergency exits
- Locations of overhead cranes, forklifts, and other material handling equipment

This layout is drawn using AutoCAD or a simplified scaled diagram, with each storage zone and rack row numbered and labelled. The storage location codes in SAP are mapped to the physical rack/bin positions.

4.4.2 Step 2: SAP Movement Frequency Analysis

Transaction MB51 is used to extract a list of all material documents (goods movements) for the last 12 months. The data fields extracted include:

- Material Number and Description
- Movement Type (GI to production, GI to maintenance, GR from vendor, etc.)
- Quantity moved
- Date of movement
- Storage Location

Using Excel pivot tables on this MB51 export, each material is assigned a Movement Frequency Count (MFC) – the number of times a goods issue transaction occurred for that material in 12 months. This forms the basis of classification:

Frequency Class	MFC (Issues per Year)	% of Materials	Assigned Zone
High Frequency (HF)	≥ 24 issues / year (bi-weekly+)	~18%	Zone A (nearest)
Medium Frequency (MF)	6–23 issues / year (monthly)	~34%	Zone B (mid-store)
Low Frequency (LF)	1–5 issues / year	~30%	Zone C (far)
Non-Moving (NM)	0 issues in last 12 months	~18%	Zone C / Disposal

4.4.3 Step 3: Travel Distance Calculation

Using the existing layout drawing, the travel distance from the issue counter to each material's current storage position is measured. The weighted average travel distance (WATD) is computed as:

WATD = Σ (Distance to material × Number of issues for material) / Total number of issues

This WATD calculation is repeated for both the existing layout and the proposed new layout to quantify the improvement. A lower WATD indicates less travel time for stores personnel, improving throughput.

4.4.4 Step 4: Proposed Layout Design

Based on the frequency classification and WATD analysis, a revised layout is proposed. The key changes in the proposed layout include:

35. Relocate all HF materials (Zone A) to racks within 10–15 metres of the issue counter and receiving bay.
36. Place MF materials (Zone B) in the central zone of the stores at 15–30 metres from the issue counter.
37. Relocate NM / LF materials to the farthest racks or to a separate slow-moving bay.
38. Widen the main aisle to allow forklift/pallet truck movement for heavy materials.
39. Create a dedicated Fast-Lane Counter for small, frequently issued consumables (fasteners, O-rings, gaskets).
40. Ensure all emergency exits are clearly marked and not obstructed by material storage.
41. Provide dedicated floor markings (painted lanes) for pedestrian and forklift traffic separation.

4.5 SAP Integration with Layout Changes

A critical aspect of the stores layout improvement is ensuring that the new physical layout is accurately reflected in SAP. Failing to update SAP bin and storage location data after a physical reorganization leads to stock discrepancies, wrong material picking, and audit failures.

4.5.1 Storage Location and Bin Management in SAP

While basic SAP MM (without WM – Warehouse Management module) uses Storage Locations as the lowest level of storage identification, units with SAP WM use Warehouse Numbers, Storage Types, and Storage Bins for granular tracking.

At BHEL, the storage location assignment in Material Master (MM02 → Storage Location data) should be verified and updated to reflect the new zone assignments after layout changes. This ensures that MMBE (Stock Overview) shows materials in their correct locations.

4.5.2 Physical Inventory Post-Reorganisation

After physically relocating materials in the stores, a targeted Physical Inventory (PI) should be conducted for all moved materials to confirm that:

- SAP stock quantity matches physical count after relocation
- No materials were lost or misplaced during movement
- Storage Location assignment in Material Master is updated

SAP Transactions for post-reorganization PI:

- MI01 – Create PI document for all relocated materials
- MI04 – Enter physical count
- MI07 – Post differences (if any)

4.6 Ergonomic and Safety Improvements

4.6.1 Rack Height and Weight Limits

As part of the layout optimization, clear guidelines are established for rack usage:

Rack Level	Height from Floor	Max Weight	Recommended Material Type
Level 1 (Bottom)	Floor to 0.8 m	Heavy (>20 kg)	Heavy castings, large boxes
Level 2 (Mid-Low)	0.8 m to 1.4 m	Medium (5–20 kg)	Valves, pumps, medium-weight parts
Level 3 (Mid-High)	1.4 m to 2.0 m	Light (<5 kg)	Small hardware, instruments

Rack Level	Height from Floor	Max Weight	Recommended Material Type
Level 4 (Top)	Above 2.0 m	Very Light (<2 kg)	Consumable boxes, documents

Heavy items must always be stored at lower levels to prevent injury during manual handling and to ensure rack stability. This is aligned with BHEL's Safety, Health, and Environment (SHE) guidelines.

4.6.2 FIFO Compliance

First-In-First-Out (FIFO) is critical for perishable or shelf-life-sensitive materials (lubricants, paints, chemicals, rubber components). The proposed layout incorporates flow-through racking where possible, ensuring that materials received first are physically accessible first. This is also aligned with SAP Batch Management (if activated) which tracks expiry dates.

4.7 Expected Results

The following quantitative improvements are expected from the proposed stores layout reorganization:

- Reduction in average material picking time: 25–30%
- Reduction in weighted average travel distance: Approx. 18%
- Reduction in material handling incidents / damage: Estimated 15%
- Improvement in 5S audit score: From approx. 62% to >80%
- Reduction in 'material not found' complaints from production: Estimated 40%

4.8 Implementation Plan

Phase	Activity	Duration	Responsibility
1	Layout survey and documentation; SAP data extraction	2 weeks	Apprentice + Stores Supvr
2	Frequency analysis using MB51 data; classification	1 week	Apprentice
3	Proposed layout design and WATD computation	1 week	Apprentice + Stores Mgr
4	Approval of proposed layout by Stores Manager	3 days	Stores Manager
5	Physical relocation of materials as per new layout	4–6 weeks	Stores Team
6	SAP Storage Location / Bin update; PI verification	1 week	SAP Team + Stores

Phase	Activity	Duration	Responsibility
7	5S audit; labelling; aisle marking; documentation	1 week	SHE Dept + Stores
8	Performance measurement; post-implementation review	Ongoing (Monthly)	Stores Manager

4.9 Conclusion – Project 2

The Stores Layout Optimization project demonstrates that systematic use of SAP MM movement data (transaction MB51) can scientifically drive a physical improvement initiative in the stores. By moving from an ad-hoc, historically-accumulated storage arrangement to a frequency-based, zoned storage system, BHEL stores can achieve significant improvements in operational efficiency, safety, and space utilisation.

The project also highlights the importance of keeping SAP storage data synchronized with the physical layout. A well-organized stores with accurate SAP records is the foundation for all downstream improvements in inventory management, production supply, and audit compliance.

5. PROJECT 3: SAP MATERIAL TRACKING AND TRACEABILITY SYSTEM

Project Summary – Project 3	
Project Title	Digital Tracking of Materials in SAP-MM for Improved Traceability
Location	BHEL – Stores & Production Departments
Duration	Approx. 2–3 months
Tools Used	SAP MM (MIGO, MB1A, MB1B, MI01, MI04, MB51, MB52, ME2M, MIRO)
Primary Objective	Map and document the end-to-end material lifecycle from GRN to production issue
Expected Outcome	Comprehensive traceability documentation; improved audit compliance; process SOP

5.1 Background and Problem Statement

In a manufacturing company like BHEL, the journey of a material from the moment it is received from a vendor to the moment it is consumed in production passes through multiple stages, each recorded in SAP through a series of transactions. This journey – from Goods Receipt to Stores to Production – is known as the Material Lifecycle.

Despite SAP providing full digital traceability of every material movement, the following challenges are commonly observed in practice:

- Stores personnel are aware of only the transactions they directly perform (e.g., MIGO for GR) but lack visibility of the broader SAP transaction chain.
- During audits (internal, CAG, statutory), auditors often ask to trace a specific material batch from GRN to production, and the lack of end-to-end SAP process knowledge makes this time-consuming.
- When quality issues arise in production, there is no quick process to trace which vendor lot the defective material came from, making vendor performance evaluation difficult.
- During project execution, materials issued to multiple WBS elements are sometimes mixed, leading to incorrect cost allocation.
- Physical inventory discrepancies are difficult to investigate when movement history is not well understood.

This project aims to create a comprehensive, step-by-step documentation of the material tracking and traceability process in SAP MM at BHEL, from Purchase Order creation all the way through to

Goods Issue to Production and Physical Inventory. The output is a process SOP (Standard Operating Procedure) that can be used by stores personnel and auditors alike.

5.2 Objectives

42. To document the complete SAP transaction flow from vendor PO to material consumption in production.
43. To identify key control points (checkpoints) in the SAP process chain where traceability is ensured or can break down.
44. To create step-by-step SAP process guides (SOPs) for each stage of the material lifecycle.
45. To demonstrate how SAP reports (MB51, MB52, MMBE, ME2M) can be used to trace any material at any stage.
46. To identify gaps in the current traceability system and recommend improvements.

5.3 The SAP Material Lifecycle – Stage by Stage

5.3.1 Stage 1: Material Master Creation / Update

Before any material can be procured or stored, it must exist as a Material Master in SAP. The Material Master is the foundational record that governs how a material is bought, stored, valued, and issued. At BHEL, Material Master creation follows a defined process involving multiple departments.

Key Material Master Views relevant to Stores:

- Basic Data 1 & 2: Material description, base unit of measure, material group, weight, dimensions
- Purchasing View: Purchasing group, order unit, GR processing time
- MRP Views 1–4: MRP type, reorder point, safety stock, lot size, planned delivery time
- Storage 1 & 2: Storage conditions, maximum storage period, temperature conditions
- Warehouse Management Views: Bin type, putaway strategy (if WM active)
- Accounting View 1 & 2: Valuation class, price control (Standard or Moving Average), standard price
- Plant Data / Storage 1: Storage location, batch classification

SAP Transactions for Material Master:

- MM01 – Create Material Master
- MM02 – Change Material Master (update parameters)
- MM03 – Display Material Master
- MM60 – Materials with Low Stock Turnover (report)

5.3.2 Stage 2: Purchase Requisition (PR) Generation

A Purchase Requisition (PR) is an internal document requesting the procurement of a material. In SAP, PRs can be generated in three ways:

47. Automatically: By MRP run (MD01/MD02) when stock falls below reorder level for MRP-managed materials.
48. Semi-automatically: By production planner based on production order material components.
49. Manually: By Stores personnel using transaction ME51N when a material is required but no MRP is active.

Every PR contains the material number, required quantity, delivery date, plant, storage location, and account assignment (cost centre / project WBS). The PR serves as the formal trigger for procurement and is subject to approval workflow in SAP.

- ME51N – Create Purchase Requisition
- ME52N – Change Purchase Requisition
- ME53N – Display Purchase Requisition
- ME5A – List of Purchase Requisitions

5.3.3 Stage 3: Purchase Order (PO) Creation and Release

A Purchase Order (PO) is the formal legal document sent to a vendor committing BHEL to purchase a specified quantity of material at an agreed price and delivery schedule. In SAP, a PO is created by the Purchase Department with reference to the approved PR.

A PO in SAP MM contains:

- Header: Vendor details, payment terms, incoterms, currency
- Line Items: Material, quantity, price, delivery date, plant, storage location, account assignment
- Conditions: Taxes (GST), freight, insurance, discounts
- Messages: Output type for printing/sending PO to vendor

After creation, the PO goes through a Release Strategy in SAP – a multi-level approval workflow based on PO value and material type. Only after all approvals are completed is the PO released and communicated to the vendor.

- ME21N – Create Purchase Order
- ME22N – Change Purchase Order
- ME23N – Display Purchase Order

- ME2M – Purchase Orders by Material
- ME28 – Release Purchase Orders

5.3.4 Stage 4: Goods Receipt (GR) – MIGO

When the vendor delivers materials to BHEL, the Stores department is responsible for performing the Goods Receipt in SAP. This is one of the most critical steps in the material lifecycle, as it:

- Records the quantity and value of materials received into BHEL's possession
- Triggers an automatic financial posting (Stock account debited, GR/IR clearing account credited)
- Creates the inventory balance in the specified Storage Location in SAP
- If Quality Management (QM) is integrated, places the material in Quality Inspection stock pending QC clearance

The primary SAP transaction for GR is MIGO (Goods Movement). The process is:

50. Open MIGO → Select 'Goods Receipt' → 'Purchase Order' → Enter PO number.
51. SAP auto-populates material, quantity, and details from the PO.
52. Verify: Physical quantity delivered vs PO quantity; check for partial deliveries.
53. Enter storage location where material will be physically stored.
54. Assign batch number (if batch management active).
55. Check 'Item OK' checkbox for each line item to be posted.
56. Post the GR → SAP creates a Material Document (10xxxxxxxx) and an Accounting Document.

Transaction Code	Description	Purpose / Remark
Material Document	Unique 10-digit document number created on GR posting	Primary traceability reference for every receipt
Posting Date	Date on which the GR is posted in SAP	Used for stock-on-date reports (MB5B)
Delivery Note	Vendor's delivery challan / invoice number	Links SAP document to physical paperwork
Batch Number	Unique batch ID assigned to this lot of material	Critical for batch-level traceability
Plant & SLoc	Plant code and Storage Location where stock is posted	Determines where material appears in stock reports
Movement Type	101 for GR against PO; 105 for release from Blocked	Controls stock type and accounting entry

Transaction Code	Description	Purpose / Remark
Vendor	Vendor number – auto-populated from PO	Links material to source vendor
GR Quantity	Quantity received – may differ from PO quantity	Tolerance check: ±% configured in SAP

5.3.5 Stage 5: Quality Inspection and Stock Release

When SAP QM is integrated with MM, materials received via MIGO (Mvt 101) are initially placed in Quality Inspection (QI) stock. They are not available for production or issue until the QC team releases them. This is a critical control point for traceability.

The QC process in SAP:

57. MIGO with Mvt 101 places stock in QI stock – visible in MMBE under 'Qual. Insp.' column.
58. QC team performs physical testing / verification of the material.
59. If material passes: QA07 / QA32 – Usage Decision 'Accept' → Mvt 321 (QI to Unrestricted).
60. If material fails: Usage Decision 'Reject' → Mvt 122 (Return to Vendor) or Mvt 344 (QI to Blocked).
61. Blocked stock can be scrapped (Mvt 551) or returned after rework.

Every step of the QI process creates a Quality Notification and Inspection Lot in SAP, providing complete traceability of quality decisions for each received lot.

5.3.6 Stage 6: Transfer Posting to Line Stores (MB1B)

For large BHEL plants, materials are first received into Main Stores and then transferred to Line Stores (stores located close to the production line) as and when required. This inter-storage location transfer is called a Transfer Posting and is performed using SAP transaction MB1B.

In MIGO, this can also be done using Movement Type 311 (Transfer Posting Storage Location to Storage Location). The transfer posting:

- Reduces stock in Main Stores (e.g., Storage Location 0001)
- Increases stock in Line Stores (e.g., Storage Location 0010)
- No financial posting occurs (both SLocs belong to the same plant and valuation area)
- Creates a material document for traceability

Inter-plant transfers (between two different BHEL plants) use a Stock Transfer Order (STO) process via ME21N (PO with document type UB) and MIGO (Mvt 351 for issue, Mvt 101 for receipt at receiving plant).

5.3.7 Stage 7: Goods Issue to Production (MB1A / MIGO – Mvt 261)

The Goods Issue (GI) to Production is the transaction where materials leave the Stores and enter the Production process. This is typically performed against a Production Order (PP-linked) or a Reservation. In SAP MM, this is performed using:

- MB1A – Goods Issue (standalone, for direct GI)
- MIGO → Goods Issue → To Production Order (Mvt 261)
- MIGO → Goods Issue → To Cost Centre (Mvt 201) – for maintenance materials
- MIGO → Goods Issue → To Project / WBS (Mvt 221) – for project-specific materials

The GI transaction:

62. Reduces stock from the Stores Storage Location in SAP.
63. Triggers financial posting: Stock account credited, Production Order / Cost Centre debited.
64. Creates a material document with the issuing Stores personnel's user ID and timestamp.
65. If batch-managed, records which specific batch was issued – critical for traceability.

Pre-GI Controls:

- The Production department raises a Material Reservation (transaction MB21) or a Production Order component list, which appears as a 'Reservation' in SAP. Stores issues against this reservation, ensuring no unauthorized material withdrawals.
- MB25 – Display Reservations: Stores can view all pending material requirements from production.

5.3.8 Stage 8: Physical Inventory Verification

Physical Inventory (PI) is the process of physically counting all materials in the Stores and reconciling the count with SAP records. BHEL conducts:

- Annual Physical Inventory: Full count of all materials – mandatory for annual financial closing.
- Cycle Count: Periodic count of selected materials (e.g., A-category monthly, B-category quarterly).
- Ad-hoc PI: Conducted when discrepancies are noticed or after major material movements.

The SAP Physical Inventory process:

Step	SAP Transaction	Activity
1	MI01	Create Physical Inventory Document – specify plant, SLoc, materials to be counted
2	MI04	Enter Inventory Count – record physical count quantity for each material
3	MI20	Print List of Inventory Differences – review variances between count and SAP book stock
4	MI07	Post Inventory Differences – after approval, adjust SAP stock to match physical count
5	MI08	Enter Count – Re-count if original variance exceeds tolerance
6	MI10	Create List of Differences – summary report for management review
7	MI35	Post Zero Stock Balance – post items not counted as zero (use with caution)

5.4 End-to-End Traceability Flow Diagram

The following table summarises the complete SAP transaction chain for material traceability from PO to production consumption. This serves as a quick reference for stores personnel and auditors:

Step	Stage	SAP Transaction	Output / Traceability Link
1	Material Master	MM01/MM02	Material Master Record – unique material number
2	Purchase Requisition	ME51N / MD01	PR Document Number → links to MRP requirement or manual request
3	Purchase Order	ME21N	PO Number → reference in all downstream SAP documents
4	Goods Receipt	MIGO (Mvt 101)	Material Document 10xxxxxxxx + Accounting Document → links GR to PO
5	Quality Inspection	QA07 / QA32	Inspection Lot Number → links QC decision to GR material document
6	QI Stock Release	MIGO (Mvt 321)	Transfer from QI to Unrestricted – new material document created
7	Transfer to Line Stores	MB1B / MIGO Mvt 311	SLoc-to-SLoc transfer material document – no accounting entry
8	Goods Issue to Prod	MIGO (Mvt 261)	GI Material Document → links consumption to Production Order / WBS
9	Physical Inventory	MI01/MI04/MI07	PI Document → reconciles physical vs book stock at period close

Step	Stage	SAP Transaction	Output / Traceability Link
10	Invoice Verification	MIRO	Accounting Document → clears GR/IR account; completes 3-way match

5.5 How to Trace a Material in SAP – Step-by-Step Guide

5.5.1 Scenario: Trace a Material from GRN to Production

Suppose a quality issue is discovered in production and the team needs to trace which vendor lot was used. The following SAP steps enable rapid traceability:

66. Identify the Production Order number or Reservation number from production records.
67. In SAP, run MB51 with the material number and plant, filtering by Movement Type 261 and the date range. This shows all GI material documents for that material.
68. Open the relevant GI material document in MB51 – note the Batch Number (if batch management is active).
69. Use MB56 (Batch Where-Used List) or MBGR (Goods Receipt with Batch) to trace the batch back to the original GR.
70. From the GR material document (Mvt 101), note the Purchase Order number.
71. Open the PO in ME23N – identify the Vendor, PO date, and quantity.
72. Check the Delivery Note / Vendor Invoice number from the GR document – this links to the physical paperwork.
73. Optionally, check MIRO for the Invoice Document to confirm the financial chain.

5.5.2 Scenario: Audit Trail for a Specific Material in Stores

During an audit, an auditor asks to show the complete transaction history for Material XXXXX at BHEL. The following SAP process provides a complete audit trail:

- MB52: Show current stock balance by Storage Location.
- MB51: Show all material documents (GR, GI, Transfer) for the last 12 or 24 months.
- MB5B: Show stock balance on any specific date in the past (e.g., year-end).
- ME2M: Show all Purchase Orders for the material.
- MB25: Show all pending Reservations for the material.
- MMBE: Show current stock overview across all levels.
- MM03: Show Material Master data including price history.

This complete audit trail from MMBE / MB51 / ME2M / MB52 demonstrates that SAP provides a fully digital, time-stamped record of every material movement, making BHEL's stores operations fully auditable and transparent.

5.6 Identified Gaps and Recommendations

5.6.1 Gap 1: Incomplete Material Master Data

Analysis of the Material Master database revealed that a significant portion of materials had incomplete data in key views – particularly missing MRP parameters, incorrect storage conditions, and outdated price data. This leads to incorrect MRP outputs and valuation errors.

Recommendation: Conduct a Material Master Data Quality audit using SAP Mass Processing (MASS transaction) to identify and correct incomplete records. Establish a process of mandatory Material Master review before annual stock-taking.

5.6.2 Gap 2: Goods Receipts Without Reference to PO

Some Goods Receipts were being posted using Mvt 501 / 521 (GR without PO reference) for emergency procurements. This breaks the procurement traceability chain as there is no PO linking the receipt to vendor, price, or approval.

Recommendation: Restrict free GR movement types (501, 521) to exceptional cases only, with mandatory approval from the Stores Manager and Finance before posting. All regular procurements must have a PO before GR.

5.6.3 Gap 3: Manual Physical Inventory with Errors

Physical inventory count sheets were being maintained on paper forms in some locations, with data later entered into SAP by a clerk. This introduces transcription errors and creates a time gap between physical count and SAP posting.

Recommendation: Adopt SAP-supported PI process using mobile data entry or printed bar-code scan sheets directly linked to SAP MI04 for count entry. This eliminates transcription errors and provides real-time count validation.

5.6.4 Gap 4: Delayed GR Postings

In several instances, Goods Receipts were posted in SAP 2–5 days after physical receipt of materials. This creates a discrepancy between physical stock and SAP stock during the gap period, misleading the production team about material availability.

Recommendation: Establish a Same-Day GR policy – all materials physically received on a given day must have their SAP GR posted on the same day. Track compliance using the Delivery Note date vs Posting Date in MIGO.

5.7 SOPs Created as Part of This Project

As a deliverable of this project, the following Standard Operating Procedures (SOPs) were created based on the SAP process documentation:

SOP#	SOP Title	Key SAP Transactions Covered
01	Goods Receipt Against Purchase Order	MIGO (Mvt 101), ME23N, MMBE
02	Quality Inspection Stock Release	QA07, QA32, MIGO (Mvt 321, 122, 344)
03	Transfer Posting – Main Stores to Line Stores	MB1B / MIGO (Mvt 311), MMBE
04	Goods Issue to Production Order	MIGO (Mvt 261), MB25, MB52
05	Goods Issue to Maintenance (Cost Centre)	MIGO (Mvt 201), IW32/IW41 (PM order)
06	Annual Physical Inventory – End-to-End	MI01, MI04, MI20, MI07, MI10
07	Reversal of Incorrect Goods Movements	MIGO (Cancel/Reverse), MB51 for verification
08	Material Traceability – Audit Investigation Guide	MB51, MB52, MMBE, MB5B, ME2M, MB56

5.8 Conclusion – Project 3

The Material Tracking and Traceability project demonstrates that SAP MM provides a comprehensive, end-to-end digital trail for every material from the moment it is planned for procurement to the moment it is consumed in production. When properly used, SAP MM eliminates the need for manual registers, reduces errors in material accounting, and provides instant audit-ready documentation.

The key value of this project is not just the technical documentation of SAP transactions, but the identification of practical gaps in how SAP is currently used and the provision of actionable recommendations to close those gaps. By standardizing the GR posting process, enforcing mandatory audit trails, and training Stores personnel on end-to-end SAP transaction flows, BHEL can significantly improve its stores compliance posture and reduce audit observations.

6. COMPARATIVE SUMMARY OF ALL THREE PROJECTS

The following table provides a consolidated comparison of the three projects undertaken during the apprenticeship:

Parameter	Project 1: Inventory Optimisation	Project 2: Layout Optimisation	Project 3: Traceability System
Focus Area	Stock levels & capital efficiency	Physical store layout & ergonomics	Digital audit trail & SOPs
Primary SAP Tool	MB52, MM60, MMBE	MB51, MMBE	MIGO, MB51, MI01–MI07
Methodology	ABC/XYZ Analysis, EOQ	Frequency mapping, WATD	Process mapping, SOP creation
Duration	3–4 months	2–3 months	2–3 months
Primary Benefit	Reduced idle inventory; working capital release	Faster material retrieval; safer stores	Audit compliance; process standardisation
Quantified Impact	₹35–50L annual savings	25–30% time reduction	Full audit trail; zero manual registers
Complexity	Medium–High	Medium	Medium–High
Interdependency	Feeds into Projects 2 & 3	Uses data from Project 1	Provides audit layer for 1 & 2

6.1 Synergy Between Projects

The three projects are not independent of each other – they form an integrated stores improvement framework:

- Project 1 (Inventory Optimisation) identifies what materials are in excess, slow-moving, or correctly stocked. This directly informs Project 2 (Layout Optimisation) about which materials should be in Zone A (fast-moving, near issue counter) vs. Zone C (slow-moving, far from counter), since the frequency classification in Project 2 is derived from the movement data analysed in Project 1.

- Project 3 (Traceability) provides the audit and compliance layer for both Project 1 and Project 2. The SOPs created in Project 3 ensure that all inventory movements, stock corrections, and PI exercises related to Project 1 and the physical reorganization in Project 2 are properly recorded in SAP with a complete audit trail.

- Together, the three projects create a virtuous cycle: Optimised inventory (Project 1) → Better organised stores (Project 2) → Accurate, traceable records (Project 3) → Enables further inventory analysis → back to Project 1.

7. LEARNING OUTCOMES FROM APPRENTICESHIP

7.1 Technical Skills Acquired

The one-year apprenticeship at BHEL's Stores and Material Management department provided the following technical competencies:

7.1.1 SAP MM Proficiency

- Hands-on use of SAP MM transactions: MIGO, MB52, MMBE, MB51, MB1A, MB1B, MM03, ME2M, MI01, MI04, MI07, MM60, MB5B
- Understanding of SAP Organizational Structure: Client, Company Code, Plant, Storage Location, Purchase Organization
- Knowledge of Movement Types and their accounting implications
- Material Master views: Purchasing, MRP, Storage, Accounting, and their inter-relationships
- SAP report extraction and data analysis using ALV Grid export to Excel
- Physical Inventory process end-to-end in SAP

7.1.2 Inventory Management Techniques

- ABC Analysis – classification by consumption value
- XYZ Analysis – classification by demand variability
- EOQ (Economic Order Quantity) computation
- Reorder Level, Safety Stock, and Maximum Stock calculation
- Inventory Turnover Ratio (ITR) interpretation and application

7.1.3 Stores Operations Knowledge

- Goods Receipt process – documentation, inspection, and SAP posting
- Material storage principles – FIFO, FEFO, weight limits, hazardous material handling
- 5S methodology implementation in an industrial stores environment
- Physical inventory verification techniques and discrepancy management
- Material handling equipment – forklifts, pallet trucks, overhead cranes – usage and safety protocols

7.2 Soft Skills and Professional Development

- Cross-functional collaboration: Working with Procurement, Production, Finance, and QC departments
- Report writing and technical documentation: Creating SOPs and project reports
- Data analysis and presentation: Converting SAP data into meaningful insights and charts
- Problem-solving in a real industrial environment under operational constraints

- Understanding of audit and compliance requirements in a PSU environment

7.3 Key Observations from the Apprenticeship

Several important observations were made during the one-year apprenticeship that are not typically covered in academic curricula:

74. SAP data quality is foundational: All analyses and reports are only as good as the underlying Material Master and transaction data. Maintaining data quality in SAP is a continuous operational discipline, not a one-time activity.
75. The human element in ERP systems: Despite sophisticated SAP configurations, many process gaps arise from users bypassing controls, posting transactions with wrong parameters, or using workaround movement types. Training and enforcement of SOPs is critical.
76. Stores is strategic, not just operational: In a ₹20,000+ crore company like BHEL, the stores function directly impacts working capital, production continuity, and financial accuracy. Stores managers need both operational skills and financial acumen.
77. Integration is SAP's greatest strength: The tight integration between MM, FI, CO, PP, and QM means that every stores transaction has ripple effects across multiple departments. Understanding these integrations is essential for effective SAP use.
78. Physical verification remains critical: Even in a fully SAP-driven environment, periodic physical verification is irreplaceable. SAP records what was posted – not necessarily what physically exists. Discrepancies between physical stock and SAP are a permanent operational reality.

8. OVERALL CONCLUSION

This apprenticeship project report presents three comprehensive studies undertaken during a one-year training programme in the Stores and Material Management department of Bharat Heavy Electricals Limited (BHEL). Each project was selected to address real and observable operational challenges in the stores function and to demonstrate how SAP MM – when used to its full potential – can drive significant improvements in inventory efficiency, operational ergonomics, and process traceability.

Project 1 – SAP-Based Inventory Optimisation – established a data-driven framework for identifying and addressing excess, slow-moving, and non-moving inventory. The ABC/XYZ analysis, combined with SAP's powerful stock reports (MB52, MM60), provides stores management with clear, actionable intelligence to reduce idle capital and prevent production stock-outs. The recommended practices, if institutionalised, can yield annual savings of ₹35–50 Lakhs in reduced inventory carrying costs and improved stock turnover.

Project 2 – Stores Layout Optimisation – demonstrated that the physical organisation of the stores space can be scientifically optimised using SAP material movement frequency data. By classifying materials into high, medium, and low frequency categories and designing a zone-based layout accordingly, the stores can reduce material picking time by 25–30%, improve safety compliance, and achieve a significant improvement in 5S audit scores. The project also highlighted the critical importance of keeping SAP storage location data synchronised with physical layout changes.

Project 3 – SAP Material Tracking and Traceability System – created a comprehensive documentation of the SAP material lifecycle from Purchase Requisition to Physical Inventory, filling a critical knowledge gap among stores personnel. The end-to-end transaction flow documentation, combined with the Standard Operating Procedures (SOPs) created as project deliverables, equips the stores team with the tools to provide instant, accurate responses to audit queries and quality investigations. The identified gaps – incomplete Material Master data, free GR postings, delayed receipts, and manual PI processes – provide a clear improvement roadmap.

Together, the three projects form a holistic Stores Excellence Framework that addresses the three dimensions of stores management: What to stock (Inventory Optimisation), Where to stock it (Layout Optimisation), and How to track it (Traceability System). This integrated approach, supported by SAP MM, positions BHEL's stores function as a competitive advantage rather than a cost centre.

The apprenticeship at BHEL has been a transformative learning experience, providing deep domain expertise in SAP MM, inventory management, and industrial stores operations that forms a strong foundation for a career in Supply Chain, Materials Management, or ERP consulting in the manufacturing sector.

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10. APPENDICES

Appendix A: Glossary of Terms

Term / Abbreviation	Definition
ABC Analysis	Classification of materials by annual consumption value (A=high, B=medium, C=low)
ALV	SAP ABAP List Viewer – a standard report output format in SAP that can be exported to Excel
BOM	Bill of Materials – a list of components required to manufacture a product
Blocked Stock	Stock received but not yet accepted into unrestricted use (pending inspection or decision)
Consignment Stock	Material physically at BHEL but owned by the vendor until consumed
EOQ	Economic Order Quantity – optimal order size minimising total ordering + holding cost
FEFO	First-Expired-First-Out – issuing material with nearest expiry date first
FIFO	First-In-First-Out – issuing material received earliest first
GI	Goods Issue – movement of material out of stores to production or other receivers
GR	Goods Receipt – movement of material into stores from vendor or other source
GR/IR	Goods Receipt / Invoice Receipt – clearing account used in 3-way matching
ITR	Inventory Turnover Ratio – annual consumption / average inventory; measures stock efficiency
LOT	A quantity of material produced or received under uniform conditions at a specific time
MRP	Material Requirements Planning – automated planning of procurement based on demand
PO	Purchase Order – formal document committing to buy from a vendor
PR	Purchase Requisition – internal request to procure a material
PSU	Public Sector Undertaking – government-owned enterprise (BHEL is a PSU)
QI Stock	Quality Inspection Stock – material pending QC clearance; not available for production
ROP	Reorder Point – stock level at which a new order should be triggered
Safety Stock	Buffer stock held to protect against demand or supply uncertainty

Term / Abbreviation	Definition
SKU	Stock Keeping Unit – a unique identifier for each distinct product / material
SLoc	Storage Location – specific physical location within a plant in SAP
STO	Stock Transfer Order – SAP document for inter-plant stock transfers
WATD	Weighted Average Travel Distance – measure used in stores layout efficiency calculation
WBS	Work Breakdown Structure – project accounting element in SAP for cost assignment
XYZ Analysis	Classification of materials by consumption variability (X=stable, Y=moderate, Z=irregular)

Appendix B: SAP MM – Complete Transaction Code Reference

Transaction Code	Description	Purpose / Remark
MB01	Post Goods Receipt (legacy)	Older GR posting transaction – use MIGO instead
MB0A	Post Goods Receipt – Batch Input	Batch input for mass GR posting
MB11	Post Goods Movement	Alternate GI/GR posting entry
MB21	Create Material Reservation	Production/maintenance material reservation
MB22	Change Material Reservation	Modify existing reservation
MB23	Display Material Reservation	View reservation details
MB24	Reservation List	List all reservations by various criteria
MB25	Reservations by MvmtType	View reservations filtered by movement type
MB31	Post Goods Receipt for Prod Order	GR for manufactured goods from production
MB51	Material Document List	Movement history of all materials
MB52	Warehouse Stocks of Material	Current unrestricted stock – key inventory report
MB53	Plant Availability	Compare stock across multiple plants

Transaction Code	Description	Purpose / Remark
MB54	Consignment Stocks	Display vendor-owned consignment stock
MB56	Batch Where-Used List	Trace batch from GR through consumption
MB58	Consignment and Pipeline Stocks	Combined view of consignment and pipeline
MB5B	Stocks for Posting Date	Historical stock as of any past date
MB5L	Stock Values – Balances	Financial value of stock by account
MB90	Output Processing for Mat Docs	Reprint material document outputs
MBGR	Display Goods Movements by Movement Reason	Filter movements by reason code
MBNL	Subsequent Delivery for Del. Cost	Delivery cost re-posting
MBST	Cancel Material Document	Cancel posted material document
MBTC	Material Documents via Batch Input	Mass posting via batch input session
MIGO	Goods Movements (All Types)	Central transaction for all inventory movements
MIRO	Enter Incoming Invoice	Invoice verification – 3-way match
MM01	Create Material Master	New material creation
MM02	Change Material Master	Update material master views
MM03	Display Material Master	View material master without changes
MM04	Material Master Change Documents	Audit trail of MM changes
MM06	Flag Material Master for Deletion	Mark obsolete materials for archiving
MM17	Mass Maintenance of Material Master	Update multiple materials simultaneously
MM50	Extend Material to New Org Level	Add new plant or SLoc to existing material
MM60	Inventory Turnover	Slow-moving / non-moving materials report
MMBE	Stock Overview	Multi-level stock display – plant, SLoc, batch

Transaction Code	Description	Purpose / Remark
MMDL	Delete Material Master	Delete material master (if no documents)
ME21N	Create Purchase Order	Create new PO with all details
ME22N	Change Purchase Order	Modify existing PO
ME23N	Display Purchase Order	View PO without changes
ME28	Release Purchase Orders	Approve POs in release strategy
ME2M	POs by Material	View all POs for a given material
ME51N	Create Purchase Requisition	Raise manual PR for procurement
ME52N	Change Purchase Requisition	Modify existing PR
ME53N	Display Purchase Requisition	View PR details
ME5A	List of Purchase Requisitions	Mass list/report of PRs by criteria
MI01	Create Physical Inventory Document	Initiate PI counting document
MI04	Enter Inventory Count	Record physical count results
MI07	Post Inventory Differences	Post count adjustments to stock
MI08	Enter Count with Reference	Re-count against existing PI document
MI10	Create List of Differences	PI variance summary report for management
MI20	Print List of Inventory Differences	Printable PI difference report
MI35	Post Zero-Count Materials	Post zero count for uncounted items
MD01	Run MRP for All Products	Mass MRP run for an entire plant
MD02	Single-Item MRP	MRP run for one material/BOM
MD04	Current Stock/Req List	MRP stock requirements list

Appendix C: Key Performance Indicators (KPIs) for BHEL Stores

KPI	Formula / Measurement	Target	SAP Report
Inventory Turnover Ratio	Annual Consumption / Avg. Stock Value	> 3.5 turns	MM60 / MB5L
Non-Moving Stock %	Value of NM Stock / Total Stock Value	< 5%	MM60
GR Accuracy Rate	GR quantity = PO quantity / Total GRs	> 98%	MB51
Same-Day GR Posting Rate	GR posted on physical receipt date	> 95%	MIGO / MB51
PI Accuracy	(Materials with zero variance / Total materials counted) × 100	> 97%	MI20 / MI10
Stock-Out Incidents	Count of production stoppages due to material unavailability	< 5 / quarter	MM60 / MB52
Stores 5S Score	Monthly 5S audit score (%)	> 80%	Manual audit
GR/GI TAT (Turnaround Time)	Avg hours from material receipt to SAP posting	< 4 hours	MIGO log

Appendix D: Sample ABC Analysis Worksheet

The following illustrative sample shows how ABC analysis is computed using SAP data exported from MB51 (annual consumption) and MB52 (current stock valuation). The steps in the Excel worksheet are:

95. Export MB51 data for last 12 months → pivot by material to get annual consumption quantity per material.
96. Export Material Master price (standard / MAP) from MM03 or a mass download.
97. Compute Annual Consumption Value (ACV) = Consumption Qty × Price for each material.
98. Sort materials in descending order of ACV.
99. Compute cumulative ACV and cumulative % of total ACV.
100. Assign A to materials up to 70% cumulative ACV; B from 70% to 90%; C from 90% to 100%.

Mat. No.	Description	Ann. Qty	Price (₹)	ACV (₹ Lac)	Category
MAT-1001	Alloy Steel Forging – Turbine Rotor	48 Nos	₹3,20,000	153.6	A
MAT-2345	Silicon Steel Lamination Sheet	12,000 kg	₹850/kg	102.0	A
MAT-3421	Copper Conductor – Winding Grade	8,500 kg	₹820/kg	69.7	A
MAT-4102	SS 316 Pipe – 2 inch NB	2,200 m	₹2,200/m	48.4	B
MAT-5678	Bearing – SKF 6205	1,800 Nos	₹1,800	32.4	B
MAT-7890	M16 Hex Bolt SS – Pack of 100	420 Packs	₹180/pk	0.76	C
MAT-8901	PVC Insulation Tape – Roll	3,600 Rolls	₹45/roll	0.16	C

Appendix E: Physical Inventory Checklist

The following checklist is used before commencing a Physical Inventory exercise at BHEL Stores:

- Obtain PI authorization from Stores Manager and Finance Manager.
- Freeze stock movements for the counted area during counting period (no GI/GR during count).
- Print count sheets from SAP (MI01) or use mobile scanning device.
- Brief counting teams on procedures: count only – do not move materials.
- Assign independent counters and supervisors to each zone.
- Record count including bin number, material code, quantity, and unit.

- Submit count sheets to SAP data entry team for MI04 entry within 2 hours of count.
- Review MI20 differences report before posting – investigate variances > 5% or > ₹10,000 value.
- Get Finance concurrence before posting MI07 for high-value differences.
- Retain count sheets and MI20 printout for audit records.
- Conduct mandatory re-count if discrepancy > 2% of total stock value.

Appendix F: SAP MM Integration Map

SAP MM integrates with the following SAP modules at BHEL, and the key integration touchpoints are:

Integrating Module	Key Integration Point	Business Impact
FI – Finance	Every GR/GI posts to GL	Stock value in balance sheet updated automatically; no manual journal
CO – Controlling	GI posts to Cost Centre / WBS	Production and project costs captured automatically from Stores issues
PP – Production Planning	BOM, Production Order, Reservation	Material requirements from production orders drive MRP and reservations in Stores
QM – Quality Mgmt	Inspection Lot, Usage Decision	QI stock control ensures no untested material issued to production
PM – Plant Maintenance	PM Work Order → GI Mvt 261	Spare parts issued against PM orders for traceability of maintenance costs
SD – Sales & Dist.	STO (Stock Transfer Order)	Inter-company or inter-plant material transfers use SD document type in STO process
WM – Warehouse Mgmt	Transfer Order, Bin Management	If WM active: bin-level traceability, putaway and pick strategies, barcode integration